Green Team 9 (B) Submission for: Daniel Yang, Philip Ratnamsamy, Ojaswi Atrey, Haider Abbas.

1. This question is about the CAPM.

(a) What is the exact definition of the beta, β, of a stock?

Beta: is a measure of stock risk related to the overall market

(b) In practice, we run a regression to compute β. Explain precisely which regression you would run to compute the beta of a stock.

$r_{i,t} - r_{f,t} = \alpha_i + \beta i\ (r_{M,t} - r_{f,t}) + \varepsilon_i$

We need to run a linear regression because CAPM assumes linear relationship between portfolio and market

(c) The CAPM says that stocks with higher betas should earn higher average returns because they are riskier. Explain carefully the intuition for why stocks with higher betas are riskier.

Because the stock with higher betas is more sensitive to market movement, and increase sensitivity means the stock with explore large fluctuations compared to market. Using $R_i=R_f+\beta(R_m-R_f)$ as reference, the term $\beta(R_m-R_f)$ becomes larger, the expected return will be higher.

2. The following diagram, which is being used to figure out the beta of a stock, is a scatter plot of excess returns on the stock against excess returns on the market. The straight line is the best-fit line through the points.

(a) Explain what is meant here by "excess" returns.

Excess return on the market = R_m-R_f

The return on the market above the risk free rate, which means the amount asset has outperformed or underperformed the risk free rate.

(b) Use the diagram to obtain an estimate of the stock's beta. Is there any evidence from the diagram that the historical average return on the stock was different from what the CAPM would have predicted? Indicate what this evidence is.

(0.31-0.05)/(0.4-0)=0.65

The historical average return on the stock was different from what the CAPM have predicted, and many of the data dots for excess return are above the CAPM line.

(c) When we studied risk-reward analysis, we measured risk using standard deviation S, or variance S 2 . When we studied the CAPM, risk was measured using something different, namely beta, β. When you are thinking about the risk of a stock, does one of the two measures make more sense than the other? Explain.

Using Standard deviation or variance will make more sense when dealing with individual stock selection since it will reflect both market and firm specific risk. Beta will make more sense when dealing with portfolio since it will have better overall measurement of how they perform together in the portfolio.

Question 3

Asset	**E(r)**	**S**	β
A	9%	10%	1
B	4%	8%	0
C	11%	14%	1.4
Market	9%	10%	
T-Bills	4%	0%	

a)

$E(r_i) - r_f = \beta i\ (E(r_M) - r_f)$

Asset A: 9% – 4% = βi (9% – 4%)

$\beta_A = 1$

Asset B: 4%– 4% = βi (9% – 4%)

$\beta_B = 0$

Asset B: 11%– 4% = βi (9% – 4%)

$\beta_C = 1.4$

b)

Systematic Risk = $\beta * S_m$

Idiosyncratic Risk = S – Systematic Risk

Asset A: Systematic Risk = 1*10% = 10% Idiosyncratic Risk = 10%-10% = **0%**

Asset B: Systematic Risk = 0*10% = 0% Idiosyncratic Risk = 8%-0% = **8%**

Asset C: Systematic Risk = 1.4*10% = 14% Idiosyncratic Risk = 14%-14% = **0%**

Asset B has the most idiosyncratic risk.